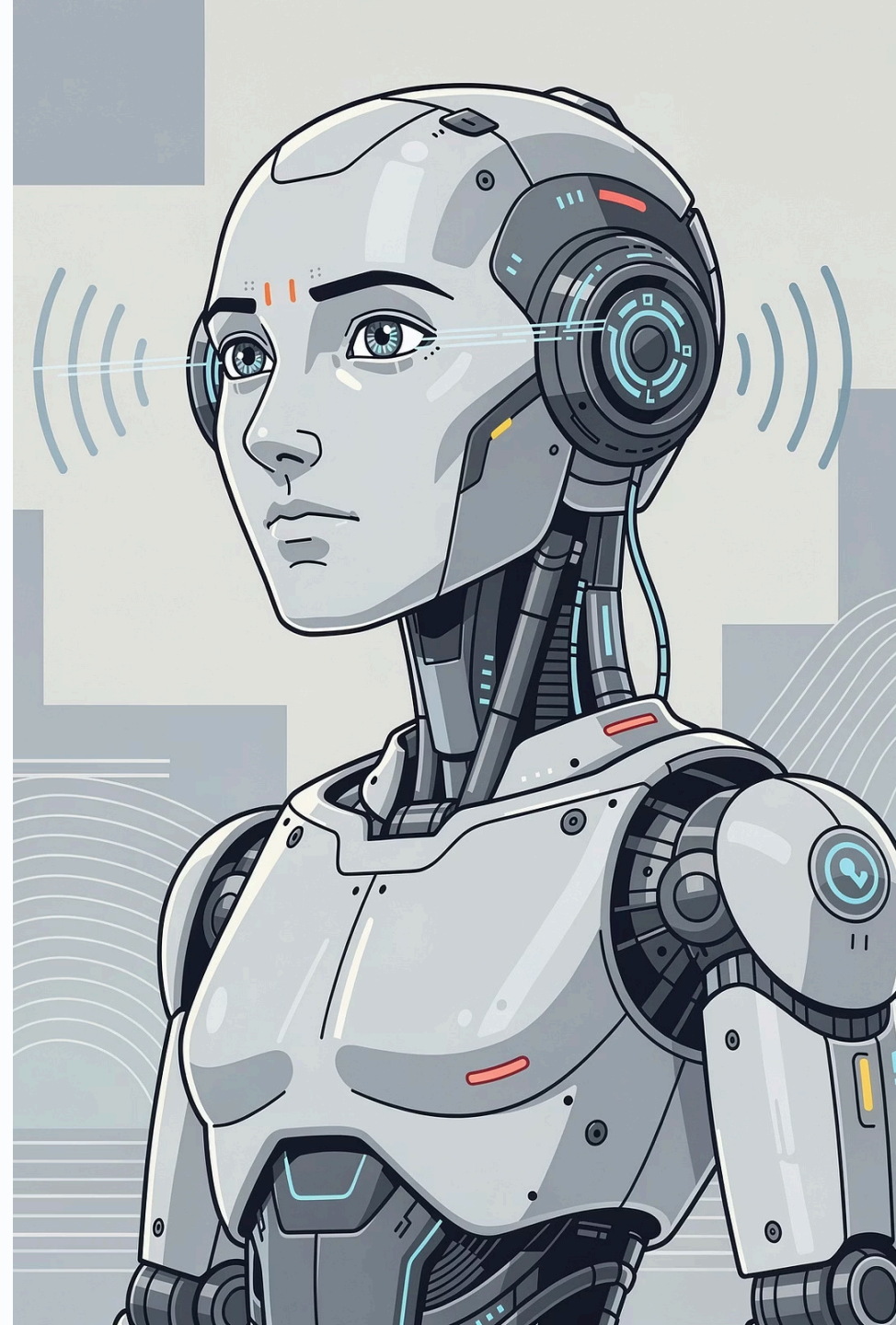


Session 17: AI That Combines Everything - Multimodal AI

You've seen AI **read**. You've seen AI **see**. You've seen AI **hear** and **speak**.

What if AI could do **ALL of those at once**? Welcome to **Multimodal AI**.



Objectives

By the end of this Session, you'll be able to:

01

Define Multimodal AI

Explain what Multimodal AI is in your own words

03

See AI Combine Senses

Discover how AI combines senses to do amazing things

05

Use Gamma

Turn your words into a full slideshow presentation

02

Know Modalities

Understand what a modality is — words, pictures, sounds, video

04

Use Leonardo

Turn your words into a video with Canva's Leonardo

06

Connect the Framework

Link back to the Sense → Think → Act framework from Session 1



Recap & Homework

Welcome back! Let's see your **Voice AI Detective notes**.

Share Your Findings

- Where did you spot Voice AI?
- Did your hands-on make a fun AI voice? Show me!
- Did any Voice AI mishear you?

Quick Check from Module 5

- NLP recipe? *Break → Understand → Generate!*
- CV recipe? *See → Process → Act!*
- Voice AI recipe? *Hear → Process → Speak!*

Warm-Up

How Do YOU Use Your Senses?

Right now, look around. What senses are you using?



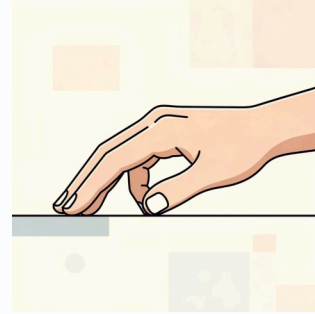
SEE

You see the screen



HEAR

You hear through your speakers



FEEL

You feel the keyboard or table



SMELL

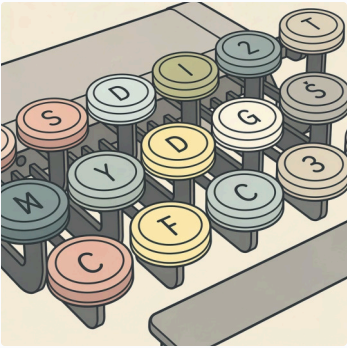
You smell flowers

① You use ALL your senses TOGETHER. You don't just "see." You see + hear + feel - all at once.

Big question: What if AI could do that too?

What's a "Modality"?

Modality is a TYPE of input or output. Don't be scared by the word - it just means a "kind of sense" for AI.



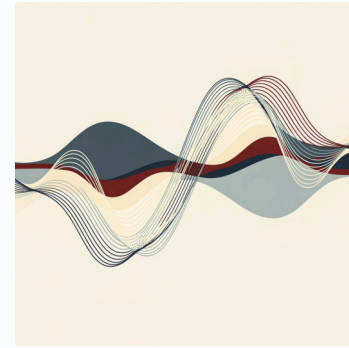
Text

Words AI **reads or writes**.



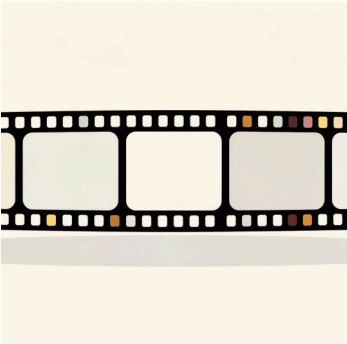
Image

Pictures AI **sees or makes**.



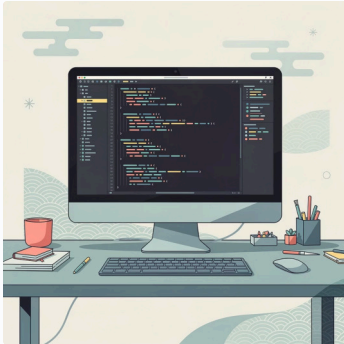
Audio

Sounds AI **hears or speaks**.



Video

Moving images AI **watches or creates**.



Code

Programming AI **reads or writes**.

What is Multimodal AI?

Multimodal AI = AI that uses MORE THAN ONE modality at once.

"Multi" = many. So "multimodal" = many types of input/output.

Text → Image

Write a story prompt, get a picture

Image → Text

Show a photo, AI describes it

Video → Text

YouTube auto-summaries

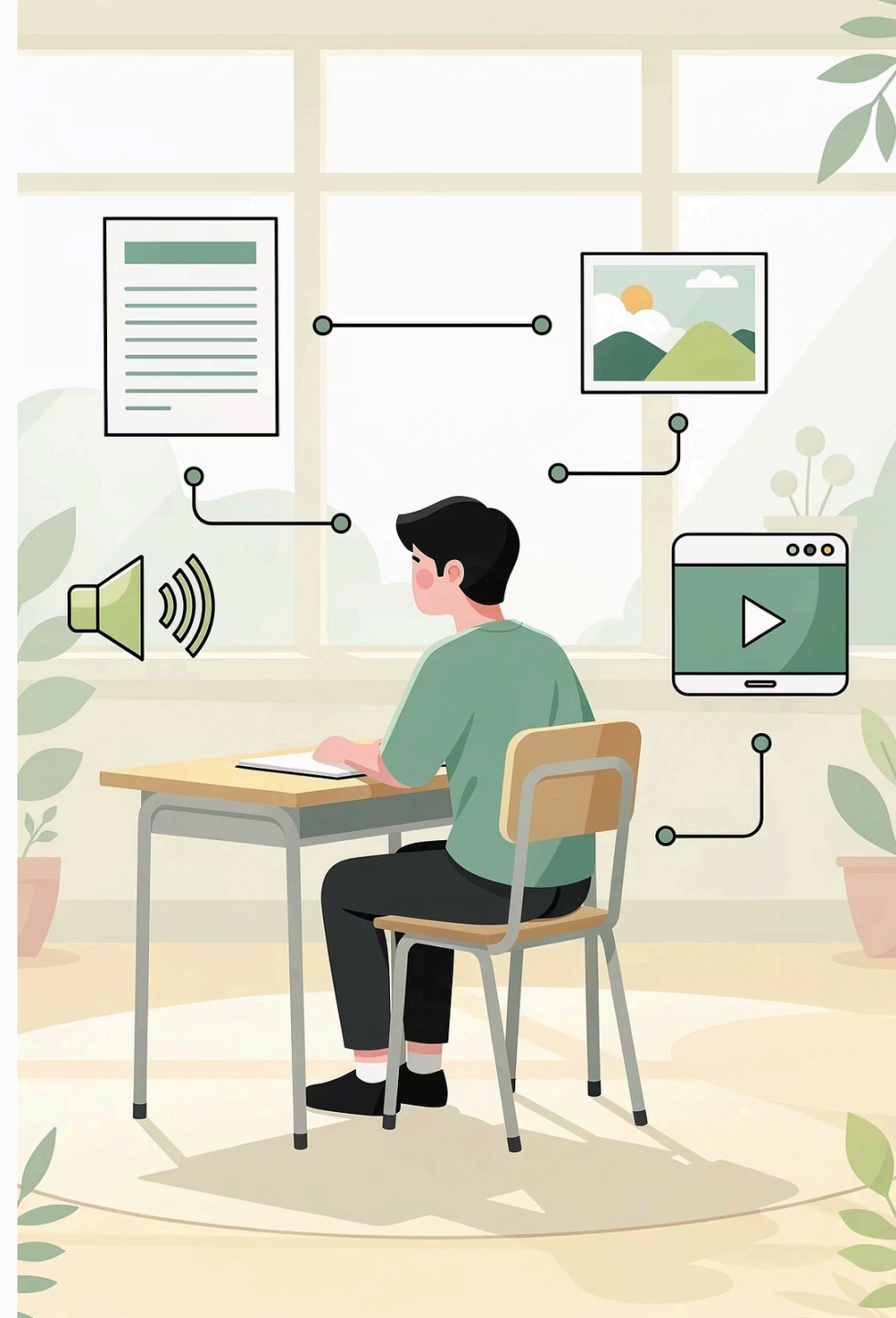
Text → Video

What we'll do today!

All Three Together

Real magic - all modalities at once

✓ **Multimodal = AI with multiple senses, all working at once.**



Remember

Sense → Think → Act?

We learned: **ALL AI follows this pattern** and Multimodal AI takes it to the next level.

AI Type	Sense	Think	Act
 NLP	Break (words)	Understand	Generate
 CV	See (pixels)	Process	Act
 Voice	Hear (sound)	Process	Speak
 Multimodal	ALL of the above	All at once	Multiple ways

Mini Activity 1 - Spot the Modalities!

For each example, name the modalities going **IN** and **OUT**. Options:  Text /  Image /  Audio /  Video

What's happening	IN (what AI receives)	OUT (what AI makes)
You type "draw a dragon" → image appears	 Text	 Image
You upload a photo → AI describes it in words	?	?
You say "make a video of a robot dancing" → video plays	?	?
You record yourself speaking English → AI gives back a video where you say it in Spanish, lips matched	?	?
AI watches a soccer game → writes a summary article	?	?

Mini Activity 1 - Answers

What's happening	IN	OUT
"Draw a dragon" → image	 Text	 Image
Upload photo → AI describes it	 Image	 Text
"Make a robot dance video" → video	 Text	 Video
Speaking English on video → Spanish with lip-sync	 Audio +  Video	 Audio (Spanish) +  Video
AI watches game → summary article	 Video	 Text

- 👉 **See the magic?** Different modalities IN, different OUT. Sometimes both. The MORE modalities AI can combine, the MORE useful and powerful it gets.

Why Multimodal AI is a BIG Deal

Before Multimodal AI

You had **separate tools** that couldn't talk to each other:

- One AI for text
- One AI for pictures
- One AI for voice
- They couldn't really "talk" to each other

With Multimodal AI

One AI can do it ALL:

- Look at your photo
- Read your question
- Think about both at once
- Answer with words OR pictures OR a video

Multimodal Examples You've Already Used

You've been using multimodal AI **without knowing it**



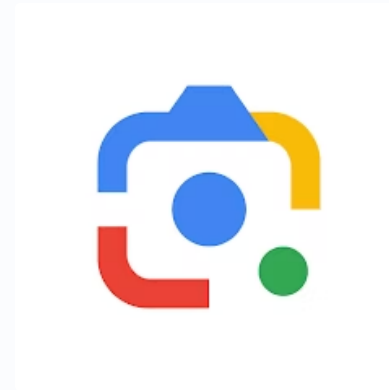
Snapchat Filters

Camera (vision) + tracking your face (vision) + voice prompts (voice)



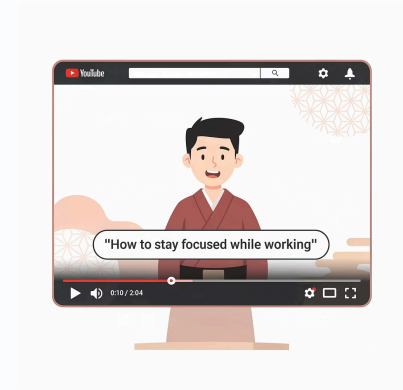
Siri Search Results

Voice (you talking) + text (search results) + sometimes images



Google Lens

Camera + text - translates signs in real-time



YouTube Auto-Captions

Audio (the voice in the video) + text (captions on screen)



The Magic of Text → Video

One of the coolest multimodal jobs: **turning a sentence into a video.**

You type: *"A cat playing piano on a rooftop at sunset."*



Subject

What does a cat look like?



Object

What does a piano look like?



Lighting

What does sunset lighting look like?



Motion

How do these things move and combine?



AI generates: 🎬 **a real video of exactly that scene!** All from one sentence. 🤖

Meet Leonardo

[Leonardo.Ai](#) is a powerful, creator-first generative AI platform designed for producing professional-quality images, textures, and videos.

It makes you a video.



Generate Video Clips

Short video clips from text prompts



Make Objects Move

Realistic motion and animation



Lighting & Atmosphere

Add lighting, atmosphere, and camera angles



Matching Audio

Sometimes adds matching audio — music or background noises!

Mini Activity 2 - Plan Your Video!

Before we generate, let's plan a great video prompt.



Action

What's happening in the video?



Subject

Who or what is in the video?



Setting

Where is it taking place?



Style

Cartoon? Realistic? Funny?



Details

Lighting, mood, camera angle

 **Try this template:** "[Subject] is [doing action] in [setting]. [Style + details]."

Example: "A golden retriever is surfing on a giant pizza in the ocean. Cartoon style, bright sunny day, happy mood."

Sample Prompts to Try

Pick one - or write your own! Your prompt should be **SPECIFIC** and **DESCRIPTIVE**.

“

Dragon Teacher

"A friendly dragon is teaching a class of children. Cartoon style, magical school, warm lighting."

”

“

Moon Dancer

"An astronaut is dancing on the moon. Realistic style, stars in the background, slow-motion."

”

“

Chess Match

"A T-Rex is playing chess against a robot. Funny cartoon, dim room, dramatic chess music."

”

“

Whale at Sunset

"A whale is leaping out of the ocean at sunset. Cinematic, glowing orange sky, slow motion."

”

“

DJ Cat

"A cat is wearing sunglasses and DJing at a beach party. Cartoon style, palm trees, party lights."

”

Hands-On Activity Part 1 - Make Your Video!

Time to generate.

01

Open [Leonardo.Ai](https://leonardo.ai)

Go to [https://leonardo.ai/](https://leonardo.ai) in the browser

02

Paste Your Prompt

Use the prompt you planned in the last activity

03

Click Generate

Hit the button and let the AI work its magic

04

Wait 30–60 Seconds...

AI is generating thousands of frames just for you!

05

Watch the Video! 🎉

Did the AI make what you imagined? Or surprise you?

❗ What did it nail? What did it get wrong? If you tweaked the prompt - what would you change?

What Just Happened?

Behind the scenes, **Leonardo did something WILD**. It combined everything you've learned in Module 5:



Read Your Prompt



Understood Every Par



Generated Pixels



Strung Frames Together



Added Matching Sounds

- ☑ That's **ALL THREE modalities from Module 5 working together.** 🤖 You just used the most multimodal AI in the course.

Meet Gamma - Multimodal for Slides! 🎨

Gamma is another multimodal AI - but for SLIDES and PRESENTATIONS. You give it a topic. It makes an entire slideshow.



Text for Slides

Writes all the content automatically



Images & Icons

Finds or generates visuals for each slide



Layout & Design

Picks a beautiful, polished layout



Full Presentation

One prompt → an entire slideshow

Hands-On Activity Part 2 - Make Your Slides!

Steps to Follow

1. Open **gamma.app**
2. Click "New" → "Generate"
3. Type a topic — anything you want!
4. Watch Gamma build you a complete slideshow

In 30 seconds, you'll have a full presentation. 🤖

Topic Ideas to Try

- 📖 "Top 5 facts about dolphins for a school project"
- 🎮 "The history of Minecraft, made for kids"
- 🐉 "A presentation on mythical creatures"
- 🌍 "Climate change explained for 5th graders"

📄 Or pick your own topic — something you're passionate about!

Compare Leonardo and Gamma

Two different multimodal tools. Both impressive in different ways. **Different multimodal jobs. Different tools. Same big idea.**

Feature	Leonardo 🎬	Gamma 🎨
Input	Text prompt	Topic / outline
Output	Short video	Full slideshow
Best for	Storytelling, creative scenes	School projects, presentations
Modalities mixed	Text → Video + Audio	Text → Text + Images + Design
Speed	30–60 seconds	~30 seconds

Reflection - What Did You Notice?

Multimodal AI is powerful. The more powerful, the more careful we have to be.

Discussion Questions

- Which tool did you like better - Leonardo or Gamma?
- What surprised you about each one?
- What did AI get RIGHT? What did it get WRONG?

Bigger Questions

✨ **Cool side:** Where could you use these for school? Or for fun? Or for storytelling?

⚠️ **Careful side:** What if someone made a FAKE video of someone real? Could it fool people?

What We Learned in this Session

You just used the most advanced AI in the entire course. 🤖 Here's everything you now know:

Modality

A type of input or output - text, image, audio, video

Multimodal AI

AI that uses more than one modality at once

Sense → Think → Act

The unifying recipe for ALL AI

Leonardo

Text → video - and you made one!

Gamma

Topic → full slideshow - and you made one!

Your Homework - Multimodal Detective 🔍

Find Multimodal AI in your daily life this week. **Mission:** Spot **3 examples** of AI using MORE THAN ONE modality.

Examples to Look For

- 📷 Snapchat or Instagram filters (vision + sometimes audio)
- 🎥 YouTube auto-captions (audio → text)
- 🌐 Google Lens (camera + text recognition)
- 📱 ChatGPT with image upload (image + text)
- 🎬 AI-generated videos online
- 🎨 Canva or Gamma making slides from a prompt

What	Modalities IN	Modalities OUT
Example 1		
Example 2		
Example 3		

Challenge

Want a stretch? **Combine TWO modalities into one creation using Canva!**

Write a Story

Write a 3-sentence story using NLP skills from Session 14

Generate an Image

Create an image for your story using Gemini

Add Narration

Generate narration using ElevenLabs (Voice, Session 16)

Generate a Video

Create a Leonardo video (Multimodal, today!)

Story + Narration (ElevenLabs)

= Audiobook 

Story + Video (Leonardo)

= Mini-Movie 

Image + Narration (ElevenLabs)

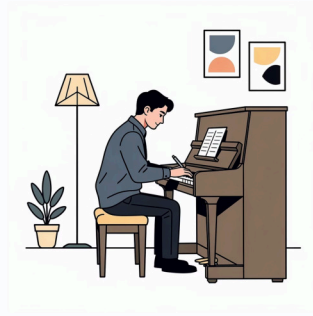
= Narrated Illustration  

Next Time: Session 18

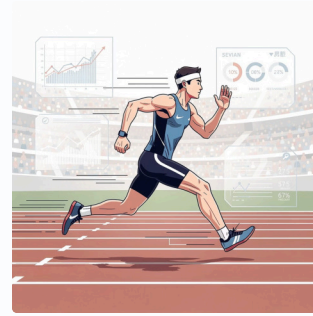
AI in Everyday Life & Across Industries



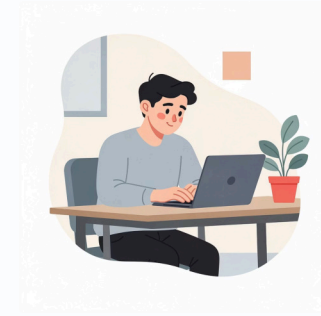
Farming



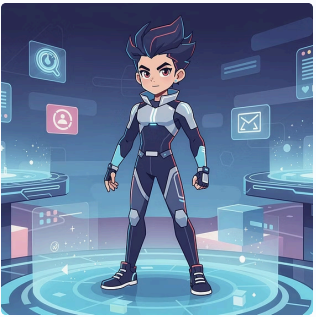
Music



Sports



Education



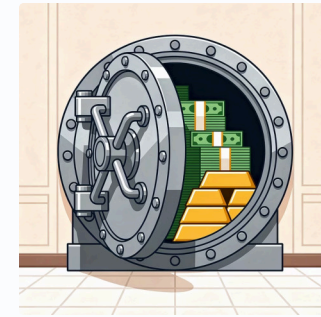
Games



Transportation



Medicine



Finance